Inflation Dynamics - Computational Notebook

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Abstract

This computational notebook documents the data construction, model specifications, estimation workflows, and best-model reporting used to study U.S. quarterly inflation dynamics.

This notebook is organized around five pieces.

1. **Data summary.** The data section constructs the effective inflation sample **1969Q2–2022Q4**, with **215 observations** and give statistical summary.
2. **Mean processes.** The mean-process menu includes Constant, ARX(1,1), ARX(2,1), and ARX(2,2) specifications. These define $\hat{\pi}_t$ and therefore the residual sequence passed to volatility models.
3. **GARCH-family volatility models.** The GARCH section estimates GARCH, GJR-GARCH, and EGARCH specifications with normal, standardized Student's t, and Gaussian-mixture residuals.
4. **Regime-switching GARCH.** The regime-switching section studies state-dependent mean dynamics with normal and standardized Student's t residuals.
5. **BEGE-GARCH models.** The BEGE section estimates Constant, Symmetric, InflationDeflation, BadGood, Constant-p Full BEGE, Constant-n Full BEGE, and Full BEGE models.

Notations are defined:

- π_t: realized inflation.
- $\mathcal{F}_{t-1}$: information set available at time $t-1$.
- SPF_{t-1}: professional forecast of inflation π_t based on $\mathcal{F}_{t-1}$.
- $\hat{\pi}_t$: expected inflation of selected mean process, $E[\pi_t \mid \mathcal{F}_{t-1}]$.
- u_t: inflation innovation, defined as $u_t := \pi_t - \hat{\pi}_t$.
- u_t^+: positive residual component, $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$.
- u_t^-: negative residual component, $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$.

The rest of the notebook uses this notation consistently across data summaries, likelihood definitions, estimation scripts, and reported best-model equations.

Computation Notebook

Published Jun 25, 2026

1. Inflation Summary Statistics

I start with simple statistical summary by plotting inflation and professional forecast over time.

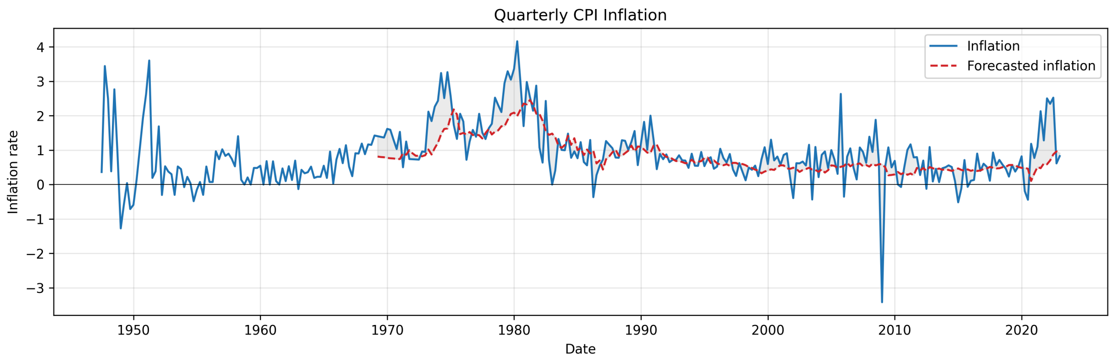


Figure 1: Quarterly inflation π_t and profesional forcast SPF_{t-1}.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Quaterly Inflation Statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use the same effective sample size for all mean models and all volatility models.[1]

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For the reported GARCH-family estimates, conditional-variance recursion starts use the default initialization implemented by the `arch` package. The estimator therefore no longer iterates the pre-sample recursion to force the initial variance to equal a parameter-implied unconditional variance.

1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

[1]Earlier versions of the estimation code contained a sample-trimming mistake and therefore used different numbers of observations across model settings. For example, some GARCH runs used 208 observations, while some BEGE runs used 210 observations. This coding error has now been corrected: all reported and regenerated estimates use the common **1969Q2–2022Q4** effective sample with **215 observations**. As a result, some current log-likelihood, AIC, and BIC values differ from earlier archived numbers.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0681	0.3482	−0.7745
P25	0.5328	0.4748	−0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	−0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + u_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + u_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + u_{t+1}$
- **ARX(2,2)** : $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_{t+1}$

where u_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{1}$$

No estimated coefficients. Residuals $u_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: −**222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: −**0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \; \underset{(0.112)}{+0.3005\,\pi_t} \; \underset{(0.167)}{+0.7337\,SPF_t} \tag{2}$$

- Log-likelihood: −**207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: −**1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \; \underset{(0.098)}{+0.2892\,\pi_t} \; \underset{(0.126)}{+0.0834\,\pi_{t-1}} \; \underset{(0.251)}{+0.6385\,SPF_t} \tag{3}$$

- Log-likelihood: −**206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: −**1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \; \underset{(0.106)}{+0.2992\,\pi_t} \; \underset{(0.125)}{+0.0914\,\pi_{t-1}} \; \underset{(0.433)}{+0.4084\,SPF_t} \; \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{4}$$

- Log-likelihood: −**206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: −**1.212**, excess kurtosis: **9.129**

3. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for residuals: normal, Student's t, and a two-component Gaussian mixture.

Let $\{u_t\}_{t=1}^{T}$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. I assume

$$u_t = \sigma_t\, z_t, \qquad z_t \mid \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{5}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or Gaussian mixture) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

3.a. *Volatility Recursion Specifications*

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i\, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k\, \sigma_{t-k}^2, \tag{6}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(u_{t-i}^{+}\right)^2 + \sum_{j=1}^{p} \gamma_j \left(u_{t-j}^{-}\right)^2 + \sum_{k=1}^{q} \beta_k\, \sigma_{t-k}^2, \tag{7}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}\left[\left(u_{t-i}^{+}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \mathbb{E}\left[\left(u_{t-i}^{-}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \frac{1}{2}\sigma_{t-i}^2 \tag{8}$$

, the stationarity condition is

$$\frac{1}{2}\sum_{i=1}^{p} \alpha_i + \frac{1}{2}\sum_{j=1}^{p} \gamma_j + \sum_{k=1}^{q} \beta_k < 1. \tag{9}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(\mid z_{t-i} \mid - \sqrt{\frac{2}{\pi}} \right) + \sum_{j=1}^{p} \gamma_j\, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{10}$$

where $z_t = u_t / \sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $(\alpha_i, \gamma_j, \beta_k)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{11}$$

3.b. Estimation Initialization And Collection

The GARCH-family estimation is executed by the `arch` package. For each distribution, mean process, volatility family, and order, the script tries the package default optimizer start and additional randomized feasible starts. For ARX mean specifications, the mean-parameter block in randomized starts is drawn from OLS-centered intervals, $\hat{\theta}_j \pm 2\,SE\left(\hat{\theta}_j\right)$, while volatility and distribution parameters are randomized over feasible regions. The current reporting run uses 50 starts per model combination. The preferred reported estimate is the highest log-likelihood estimate among starts that both converged according to the optimizer and passed the relevant stationarity proxy with a small numerical margin below one (`1e-6`). If none of the 50 starts passes all checks, the script reports the best optimizer-converged attempted log likelihood for that model combination and flags the row in the CSV `selection_status` column. Attempt-level restart diagnostics are also saved for audit. Conditional-variance recursion starts use the default initialization implemented by the `arch` package.

3.c. Log-Likelihood Specifications

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{12}$$

3.c.i. *Normal distribution:*

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0,1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{13}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t}\,\phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\,\sigma_t^2}} \exp\left\{-\frac{u_t^2}{2\,\sigma_t^2}\right\}. \tag{14}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{15}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{16}$$

3.c.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}}\left(1+\frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \tag{17}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\,\sigma_t^2}}\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{18}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\sigma_t^2\ \right) - \frac{\nu+1}{2}\ln\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{19}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\ \right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1+\frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right). \tag{20}$$

3.c.iii. *Gaussian mixture distribution:*

The standardized innovation follows a two-component Gaussian mixture. The free parameters are (p_1, μ_1, σ_1^2):

$$f_{\text{mix}}(z) = p_1\,\phi(z;\mu_1,\sigma_1^2) + p_2\,\phi(z;\mu_2,\sigma_2^2), \qquad \phi(z;\mu,s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{21}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\text{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2\mu_2^2}{p_2}. \tag{22}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\text{mix}}(a) = p_1\,\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\,\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{23}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t} f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{24}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \tag{25}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T} \ln \sigma_t + \sum_{t=1}^{T} \ln\left[p_1\, \phi(z_t; \mu_1, \sigma_1^2) + p_2\, \phi(z_t; \mu_2, \sigma_2^2) \right]. \tag{26}$$

4. Model Selection Report

For each model combination, the reported row uses the highest log-likelihood estimate that passes optimizer convergence and stationarity checks across 50 starts. If no start passes all checks, the reported row uses the best optimizer-converged attempted log likelihood and is flagged in the CSV `selection_status` column.

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH[2]

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	ARX(2,1)	(2,1)	387.2314	ARX(1,1)	(2,1)	372.5785	ARX(1,1)	(1,2)	358.6843
Student's t	ARX(2,1)	(1,1)	361.3528	ARX(2,1)	(2,1)	356.9953	ARX(2,1)	(1,2)	351.8931
Gaussian mixture	ARX(1,1)	(2,1)	367.1233	ARX(2,1)	(2,1)	358.4230	ARX(2,1)	(1,1)	352.7338

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	Constant	(2,1)	401.2826	Constant	(2,1)	396.7277	ARX(1,1)	(1,2)	385.6494
Student's t	Constant	(2,1)	386.6231	Constant	(1,1)	384.2364	Constant	(1,1)	383.5813
Gaussian mixture	ARX(1,1)	(2,1)	400.8297	ARX(1,1)	(1,1)	394.1515	Constant	(1,1)	385.2612

4.a. *Best Models by Criterion and Volatility Family*

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits. The main result tables contain 144 reported model combinations.

4.a.i. *AIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M073`	Student's t	ARX(2,1)	GARCH(1,1)	−172.6764	361.3528	388.3179

Mean process:

$$\hat{\pi}_{t+1} = 0.0876 + 0.2637\,\pi_t + 0.1716\,\pi_{t-1} + 0.5096\,SPF_t + \mu_{t+1} \tag{27}$$

Volatility process:

[2] These numbers differ from earlier GARCH tables because earlier code used inconsistent effective sample sizes across model settings due to a sample-trimming error. The correction is discussed in the Data Summary effective-sample section. The current estimates use the common **1969Q2–2022Q4** sample with **215 observations**.

$$\hat{\sigma}_t^2 = 0.0725 + 0.5202\, u_{t-1}^2 + 0.4294\, \sigma_{t-1}^2 \tag{28}$$

Distribution parameters:

Parameter	Estimate
ν	4.3024

Parameter	Coef	Std Err	t-value	p-value
c	0.0876	0.0777	1.1279	0.2593
ρ_1	0.2637	0.0912	2.8914	0.0038
ρ_2	0.1716	0.0975	1.7600	0.0784
ϕ_1	0.5096	0.1487	3.4274	0.0006
ω	0.0725	0.0385	1.8834	0.0596
α_1	0.5202	0.2424	2.1465	0.0318
β_1	0.4294	0.1425	3.0126	0.0026
ν	4.3024	1.4647	2.9373	0.0033

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M080`	Student's t	ARX(2,1)	GJR-GARCH(2,1)	−167.4976	356.9953	394.0723

Mean process:

$$\hat{\pi}_{t+1} = 0.0901 + 0.2045\, \pi_t + 0.1144\, \pi_{t-1} + 0.6522\, SPF_t + \mu_{t+1} \tag{29}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.1096 + 0.4005\, (u_{t-1}^+)^2 + 0.2848\, (u_{t-1}^-)^2 + 0.6415\, (u_{t-2}^+)^2 + 0.0000\, (u_{t-2}^-)^2 + 0.1559\, \sigma_{t-1}^2 \tag{30}$$

Distribution parameters:

Parameter	Estimate
ν	5.3421

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0901	0.0838	1.0747	0.2825
ρ_1	0.2045	0.0960	2.1292	0.0332
ρ_2	0.1144	0.1043	1.0969	0.2727
ϕ_1	0.6522	0.1740	3.7481	0.0002
ω	0.1096	0.0397	2.7593	0.0058

Parameter	Coef	Std Err	t-value	p-value
α_1	0.4005	0.2194	1.8250	0.0680
$\gamma_1 - \alpha_1$	−0.1157	0.2773	−0.4173	0.6765
α_2	0.6415	0.3571	1.7962	0.0725
$\gamma_2 - \alpha_2$	−0.6415	0.3194	−2.0082	0.0446
β_1	0.1559	0.1500	1.0399	0.2984
ν	5.3421	1.9529	2.7354	0.0062

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M078`	Student's t	ARX(2,1)	EGARCH(1,2)	−165.9465	351.8931	385.5994

Mean process:

$$\hat{\pi}_{t+1} = 0.0074 + 0.1430\,\pi_t + 0.1015\,\pi_{t-1} + 0.8896\,SPF_t + \mu_{t+1} \tag{31}$$

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.0740 - 0.3421\left(|\ z_{t-1}\ | - \sqrt{2/\pi}\right) + 0.2355\,z_{t-1} + 0.0412\ln\sigma_{t-1}^2 + 0.9200\ln\sigma_{t-2}^2 \tag{32}$$

Distribution parameters:

Parameter	Estimate
ν	4.5989

Parameter	Coef	Std Err	t-value	p-value
c	0.0074	0.0002	33.8794	0.0000
ρ_1	0.1430	0.0016	91.7625	0.0000
ρ_2	0.1015	0.0016	63.3776	0.0000
ϕ_1	0.8896	0.0000	2430766.0218	0.0000
ω	−0.0740	0.0000	−50401.4684	0.0000
α_1	−0.3421	0.0001	−2676.8779	0.0000
γ_1	0.2355	0.0010	241.7362	0.0000
β_1	0.0412	0.0002	186.9354	0.0000
β_2	0.9200	0.0000	728877.9846	0.0000
ν	4.5989	0.2382	19.3104	0.0000

4.a.ii. *BIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M055`	Student's t	Constant	GARCH(2,1)	−179.8850	369.7699	386.6231

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{33}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.1192 + 0.4570\, u_{t-1}^2 + 0.5209\, u_{t-2}^2 + 0.0000\, \sigma_{t-1}^2 \tag{34}$$

Distribution parameters:

Parameter	Estimate
ν	4.9416

Parameter	Coef	Std Err	t-value	p-value
ω	0.1192	0.0417	2.8561	0.0043
α_1	0.4570	0.1774	2.5755	0.0100
α_2	0.5209	0.2133	2.4419	0.0146
β_1	0.0000	0.1189	0.0000	1.0000
ν	4.9416	1.7105	2.8890	0.0039

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M050`	Student's t	Constant	GJR-GARCH(1,1)	−178.6916	367.3832	384.2364

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{35}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.0636 + 0.7969\, \left(u_{t-1}^{+}\right)^2 + 0.1469\, \left(u_{t-1}^{-}\right)^2 + 0.4353\, \sigma_{t-1}^2 \tag{36}$$

Distribution parameters:

Parameter	Estimate
ν	4.7915

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
Parameter	Coef	Std Err	t-value	p-value
ω	0.0636	0.0255	2.4924	0.0127
α_1	0.7969	0.2566	3.1059	0.0019
$\gamma_1 - \alpha_1$	−0.6499	0.2320	−2.8010	0.0051
β_1	0.4353	0.1168	3.7261	0.0002
ν	4.7915	1.5515	3.0883	0.0020

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M051`	Student's t	Constant	EGARCH(1,1)	−178.3641	366.7281	383.5813

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{37}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.2496 + 0.5580 \left(\mid z_{t-1} \mid - \sqrt{2/\pi} \right) + 0.2647\, z_{t-1} + 0.7897 \ln \sigma_{t-1}^2 \tag{38}$$

Distribution parameters:

Parameter	Estimate
ν	5.0826

Parameter	Coef	Std Err	t-value	p-value
ω	−0.2496	0.1063	−2.3491	0.0188
α_1	0.5580	0.1395	4.0011	0.0001
γ_1	0.2647	0.0782	3.3870	0.0007
β_1	0.7897	0.0610	12.9555	0.0000
ν	5.0826	1.7271	2.9428	0.0033

4.b. All Model Results

Distribution	Mean Process	Volatility Process	Best Looglik	AIC	BIC
Normal	Constant	GARCH(1,1)	−198.2015	402.4029	412.5148
Normal	Constant	GARCH(1,2)	−198.2015	404.4029	417.8855
Normal	Constant	GARCH(2,1)	−189.9000	387.8000	401.2826
Normal	Constant	GARCH(2,2)	−189.9000	389.8000	406.6532
Normal	Constant	GJR-GARCH(1,1)	−189.8927	387.7854	401.2680
Normal	Constant	GJR-GARCH(1,2)	−189.8927	389.7854	406.6386

Normal	Constant	GJR-GARCH(2,1)	−182.2520	376.5039	396.7277
Distribution	Mean Process	Volatility Process	Best Looglik	AIC	BIC
Normal	Constant	GJR-GARCH(2,2)	−182.2520	378.5039	402.0984
Normal	Constant	EGARCH(1,1)	−188.9696	385.9392	399.4217
Normal	Constant	EGARCH(1,2)	−188.9696	387.9392	404.7924
Normal	Constant	EGARCH(2,1)	−182.9516	377.9032	398.1271
Normal	Constant	EGARCH(2,2)	−182.9516	379.9032	403.4977
Normal	ARX(1,1)	GARCH(1,1)	−191.0320	394.0640	414.2879
Normal	ARX(1,1)	GARCH(1,2)	−191.0320	396.0640	419.6585
Normal	ARX(1,1)	GARCH(2,1)	−186.6387	387.2773	410.8718
Normal	ARX(1,1)	GARCH(2,2)	−186.6387	389.2773	416.2424
Normal	ARX(1,1)	GJR-GARCH(1,1)	−183.8229	381.6458	405.2403
Normal	ARX(1,1)	GJR-GARCH(1,2)	−183.8229	383.6458	410.6109
Normal	ARX(1,1)	GJR-GARCH(2,1)	−177.2892	372.5785	402.9142
Normal	ARX(1,1)	GJR-GARCH(2,2)	−177.2892	374.5785	408.2849
Normal	ARX(1,1)	EGARCH(1,1)	−183.7595	381.5190	405.1134
Normal	ARX(1,1)	EGARCH(1,2)	−171.3421	358.6843	385.6494
Normal	ARX(1,1)	EGARCH(2,1)	−178.1375	374.2750	404.6107
Normal	ARX(1,1)	EGARCH(2,2)	−171.2540	362.5081	396.2144
Normal	ARX(2,1)	GARCH(1,1)	−188.8527	391.7055	415.2999
Normal	ARX(2,1)	GARCH(1,2)	−188.8527	393.7055	420.6706
Normal	ARX(2,1)	GARCH(2,1)	−185.6157	387.2314	414.1965
Normal	ARX(2,1)	GARCH(2,2)	−185.6157	389.2314	419.5672
Normal	ARX(2,1)	GJR-GARCH(1,1)	−183.7897	383.5794	410.5445
Normal	ARX(2,1)	GJR-GARCH(1,2)	−183.7897	385.5794	415.9151
Normal	ARX(2,1)	GJR-GARCH(2,1)	−176.6378	373.2755	406.9819
Normal	ARX(2,1)	GJR-GARCH(2,2)	−176.6378	375.2755	412.3525
Normal	ARX(2,1)	EGARCH(1,1)	−183.4410	382.8820	409.8471
Normal	ARX(2,1)	EGARCH(1,2)	−171.1430	360.2860	390.6217
Normal	ARX(2,1)	EGARCH(2,1)	−177.9523	375.9046	409.6110
Normal	ARX(2,1)	EGARCH(2,2)	−177.9523	377.9046	414.9816
Normal	ARX(2,2)	GARCH(1,1)	−188.8355	393.6711	420.6362
Normal	ARX(2,2)	GARCH(1,2)	−188.8355	395.6711	426.0068
Normal	ARX(2,2)	GARCH(2,1)	−185.6074	389.2147	419.5505
Normal	ARX(2,2)	GARCH(2,2)	−185.6074	391.2147	424.9211
Normal	ARX(2,2)	GJR-GARCH(1,1)	−183.7821	385.5642	415.9000
Normal	ARX(2,2)	GJR-GARCH(1,2)	−183.7821	387.5642	421.2706

Normal	ARX(2,2)	GJR-GARCH(2,1)	−176.5660	375.1321	412.2091
Distribution	Mean Process	Volatility Process	Best Looglik	AIC	BIC
Normal	ARX(2,2)	GJR-GARCH(2,2)	−176.5660	377.1321	417.5797
Normal	ARX(2,2)	EGARCH(1,1)	−182.8681	383.7363	414.0720
Normal	ARX(2,2)	EGARCH(1,2)	−182.8681	385.7363	419.4427
Normal	ARX(2,2)	EGARCH(2,1)	−177.5298	377.0595	414.1365
Normal	ARX(2,2)	EGARCH(2,2)	−177.5298	379.0595	419.5072
Student’s t	Constant	GARCH(1,1)	−181.9942	371.9883	385.4709
Student’s t	Constant	GARCH(1,2)	−181.9942	373.9883	390.8415
Student’s t	Constant	GARCH(2,1)	−179.8850	369.7699	386.6231
Student’s t	Constant	GARCH(2,2)	−179.8850	371.7699	391.9938
Student’s t	Constant	GJR-GARCH(1,1)	−178.6916	367.3832	384.2364
Student’s t	Constant	GJR-GARCH(1,2)	−178.6916	369.3832	389.6070
Student’s t	Constant	GJR-GARCH(2,1)	−174.8160	363.6320	387.2265
Student’s t	Constant	GJR-GARCH(2,2)	−174.8160	365.6320	392.5972
Student’s t	Constant	EGARCH(1,1)	−178.3641	366.7281	383.5813
Student’s t	Constant	EGARCH(1,2)	−178.3641	368.7281	388.9520
Student’s t	Constant	EGARCH(2,1)	−175.5435	365.0870	388.6815
Student’s t	Constant	EGARCH(2,2)	−175.5435	367.0870	394.0521
Student’s t	ARX(1,1)	GARCH(1,1)	−174.9174	363.8348	387.4293
Student’s t	ARX(1,1)	GARCH(1,2)	−174.9174	365.8348	392.7999
Student’s t	ARX(1,1)	GARCH(2,1)	−173.7890	363.5780	390.5431
Student’s t	ARX(1,1)	GARCH(2,2)	−173.7890	365.5780	395.9138
Student’s t	ARX(1,1)	GJR-GARCH(1,1)	−171.2540	358.5080	385.4731
Student’s t	ARX(1,1)	GJR-GARCH(1,2)	−171.2540	360.5080	390.8437
Student’s t	ARX(1,1)	GJR-GARCH(2,1)	−168.5408	357.0815	390.7879
Student’s t	ARX(1,1)	GJR-GARCH(2,2)	−168.5408	359.0815	396.1586
Student’s t	ARX(1,1)	EGARCH(1,1)	−171.0753	358.1506	385.1157
Student’s t	ARX(1,1)	EGARCH(1,2)	−171.0753	360.1506	390.4864
Student’s t	ARX(1,1)	EGARCH(2,1)	−168.3693	356.7385	390.4449
Student’s t	ARX(1,1)	EGARCH(2,2)	−168.3693	358.7385	395.8155
Student’s t	ARX(2,1)	GARCH(1,1)	−172.6764	361.3528	388.3179
Student’s t	ARX(2,1)	GARCH(1,2)	−172.6764	363.3528	393.6885
Student’s t	ARX(2,1)	GARCH(2,1)	−171.9737	361.9474	392.2832
Student’s t	ARX(2,1)	GARCH(2,2)	−171.6977	363.3953	397.1017
Student’s t	ARX(2,1)	GJR-GARCH(1,1)	−170.0718	358.1436	388.4794
Student’s t	ARX(2,1)	GJR-GARCH(1,2)	−170.0718	360.1436	393.8500

Distribution	Mean Process	Volatility Process	Best Looglik	AIC	BIC
Student's t	ARX(2,1)	GJR-GARCH(2,1)	−167.4976	356.9953	394.0723
Student's t	ARX(2,1)	GJR-GARCH(2,2)	−167.4976	358.9953	399.4429
Student's t	ARX(2,1)	EGARCH(1,1)	−169.7106	357.4213	387.7570
Student's t	ARX(2,1)	EGARCH(1,2)	−165.9465	351.8931	385.5994
Student's t	ARX(2,1)	EGARCH(2,1)	−167.5810	357.1621	394.2391
Student's t	ARX(2,1)	EGARCH(2,2)	−167.5810	359.1621	399.6098
Student's t	ARX(2,2)	GARCH(1,1)	−172.5010	363.0019	393.3377
Student's t	ARX(2,2)	GARCH(1,2)	−172.5010	365.0019	398.7083
Student's t	ARX(2,2)	GARCH(2,1)	−171.8786	363.7573	397.4636
Student's t	ARX(2,2)	GARCH(2,2)	−171.6518	365.3035	402.3805
Student's t	ARX(2,2)	GJR-GARCH(1,1)	−169.9900	359.9800	393.6864
Student's t	ARX(2,2)	GJR-GARCH(1,2)	−169.9900	361.9800	399.0571
Student's t	ARX(2,2)	GJR-GARCH(2,1)	−167.3944	358.7887	399.2364
Student's t	ARX(2,2)	GJR-GARCH(2,2)	−167.3944	360.7887	404.6070
Student's t	ARX(2,2)	EGARCH(1,1)	−169.4442	358.8885	392.5948
Student's t	ARX(2,2)	EGARCH(1,2)	−169.4442	360.8885	397.9655
Student's t	ARX(2,2)	EGARCH(2,1)	−167.2956	358.5911	399.0388
Student's t	ARX(2,2)	EGARCH(2,2)	−167.2956	360.5911	404.4094
Gaussian mixture	Constant	GARCH(1,1)	−179.4873	370.9746	391.1984
Gaussian mixture	Constant	GARCH(1,2)	−179.4873	372.9746	396.5690
Gaussian mixture	Constant	GARCH(2,1)	−177.7161	369.4323	393.0268
Gaussian mixture	Constant	GARCH(2,2)	−177.7161	371.4323	398.3974
Gaussian mixture	Constant	GJR-GARCH(1,1)	−175.2123	364.4246	388.0190
Gaussian mixture	Constant	GJR-GARCH(1,2)	−175.2123	366.4246	393.3897
Gaussian mixture	Constant	GJR-GARCH(2,1)	−171.8210	361.6420	391.9777
Gaussian mixture	Constant	GJR-GARCH(2,2)	−171.8210	363.6420	397.3484
Gaussian mixture	Constant	EGARCH(1,1)	−173.8334	361.6667	385.2612
Gaussian mixture	Constant	EGARCH(1,2)	−173.8334	363.6667	390.6318
Gaussian mixture	Constant	EGARCH(2,1)	−170.7788	359.5577	389.8934
Gaussian mixture	Constant	EGARCH(2,2)	−170.7788	361.5577	395.2641
Gaussian mixture	ARX(1,1)	GARCH(1,1)	−174.5852	367.1704	397.5061
Gaussian mixture	ARX(1,1)	GARCH(1,2)	−174.5852	369.1704	402.8768
Gaussian mixture	ARX(1,1)	GARCH(2,1)	−173.5617	367.1233	400.8297
Gaussian mixture	ARX(1,1)	GARCH(2,2)	−173.5617	369.1233	406.2004
Gaussian mixture	ARX(1,1)	GJR-GARCH(1,1)	−170.2226	360.4452	394.1515
Gaussian mixture	ARX(1,1)	GJR-GARCH(1,2)	−170.2226	362.4452	399.5222

Distribution	Mean Process	Volatility Process	Best Looglik	AIC	BIC
Gaussian mixture	ARX(1,1)	GJR-GARCH(2,1)	−167.4469	358.8939	399.3415
Gaussian mixture	ARX(1,1)	GJR-GARCH(2,2)	−167.4469	360.8939	404.7122
Gaussian mixture	ARX(1,1)	EGARCH(1,1)	−169.4422	358.8844	392.5908
Gaussian mixture	ARX(1,1)	EGARCH(1,2)	−169.4422	360.8844	397.9614
Gaussian mixture	ARX(1,1)	EGARCH(2,1)	−166.2197	356.4394	396.8871
Gaussian mixture	ARX(1,1)	EGARCH(2,2)	−166.2197	358.4394	402.2577
Gaussian mixture	ARX(2,1)	GARCH(1,1)	−171.3839	362.7678	396.4742
Gaussian mixture	ARX(2,1)	GARCH(1,2)	−171.3839	364.7678	401.8449
Gaussian mixture	ARX(2,1)	GARCH(2,1)	−170.7809	363.5618	400.6389
Gaussian mixture	ARX(2,1)	GARCH(2,2)	−170.6186	365.2371	405.6848
Gaussian mixture	ARX(2,1)	GJR-GARCH(1,1)	−168.6131	359.2261	396.3031
Gaussian mixture	ARX(2,1)	GJR-GARCH(1,2)	−168.6131	361.2261	401.6738
Gaussian mixture	ARX(2,1)	GJR-GARCH(2,1)	−166.2115	358.4230	402.2413
Gaussian mixture	ARX(2,1)	GJR-GARCH(2,2)	−166.2115	360.4230	407.6120
Gaussian mixture	ARX(2,1)	EGARCH(1,1)	−165.3669	352.7338	389.8108
Gaussian mixture	ARX(2,1)	EGARCH(1,2)	−164.6031	353.2062	393.6539
Gaussian mixture	ARX(2,1)	EGARCH(2,1)	−165.2689	356.5377	400.3560
Gaussian mixture	ARX(2,1)	EGARCH(2,2)	−163.6882	355.3765	402.5654
Gaussian mixture	ARX(2,2)	GARCH(1,1)	−171.1994	364.3989	401.4759
Gaussian mixture	ARX(2,2)	GARCH(1,2)	−171.1994	366.3989	406.8465
Gaussian mixture	ARX(2,2)	GARCH(2,1)	−170.7011	365.4022	405.8499
Gaussian mixture	ARX(2,2)	GARCH(2,2)	−174.8814	375.7628	419.5811
Gaussian mixture	ARX(2,2)	GJR-GARCH(1,1)	−168.4801	360.9602	401.4079
Gaussian mixture	ARX(2,2)	GJR-GARCH(1,2)	−168.4801	362.9602	406.7785
Gaussian mixture	ARX(2,2)	GJR-GARCH(2,1)	−166.0585	360.1169	407.3059
Gaussian mixture	ARX(2,2)	GJR-GARCH(2,2)	−166.0585	362.1169	412.6765
Gaussian mixture	ARX(2,2)	EGARCH(1,1)	−167.3628	358.7257	399.1733
Gaussian mixture	ARX(2,2)	EGARCH(1,2)	−166.4468	358.8937	402.7120
Gaussian mixture	ARX(2,2)	EGARCH(2,1)	−162.5823	353.1646	400.3535
Gaussian mixture	ARX(2,2)	EGARCH(2,2)	−164.5054	359.0108	409.5704

5. Regime-Switching Inflation Models

Let the latent state be

$$s_t \in \{1, ..., K\}, \qquad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \tag{39}$$

I estimate several specifications that differ in **which blocks of parameters are allowed to switch with s_t**. Taking the ARX(2,2) model as an example,

$$\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 \,\mathrm{SPF}_t + \phi_2 \,\mathrm{SPF}_{t-1} + \mu_{t+1}, \tag{40}$$

the parameters are grouped into three blocks, and each block can independently be made regime-dependent:

Block	State-dependent parameters
AR component	$c_{s_t},\ \rho_{1,s_t},\ \rho_{2,s_t}$
SPF component	$\phi_{1,s_t},\ \phi_{2,s_t}$
Shock distribution	$\sigma_{s_t},\ \nu_{s_t}$

where ν denotes the degrees of freedom of the standardized Student-t residual. The shock-distribution block is treated as an indivisible unit: when it switches, both σ^2 and ν are regime-specific; when it does not switch, both are held constant across regimes. A given specification is defined by selecting any subset of these three blocks to be switching; parameters in blocks not selected are held constant across regimes.

5.a. *Estimation And Collection*

For each regime-switching specification, I run 50 optimizer starts. Each start first uses 50 EM iterations to improve the starting values and then applies MLE to fine tune the likelihood. For Student-t specifications the EM phase uses the normal-distribution M-step as an approximation — correct EM for the Student-t (ECME) would additionally require posterior scale-mixture weights $w_t = \left(\hat{\nu}_{s_t} + 1\right) / \left(\hat{\nu}_{s_t} + u_t^2 / \hat{\sigma}_{s_t}^2\right)$, which are not available in the base estimator; all ν_{s_t} are therefore frozen during EM and updated only by the subsequent MLE phase. The collected estimate for a model specification is the highest log-likelihood attempt that passes all checks:

- optimizer convergence,
- AR mean-process stationarity in every regime,
- positive non-boundary variance and interior, non-Gaussian-limit Student-t degrees of freedom when applicable,
- and valid non-boundary transition probabilities.

The collection step rejects selected attempts with variance estimates at or below `max(1e-6, 1e-3 * var(endog))`, transition probabilities within `1e-4` of 0 or 1, or Student-t degrees of freedom at or above `80`.

For the AR stationarity check, only the inflation-lag coefficients enter the AR polynomial. SPF coefficients are treated as exogenous loadings. Specifications where none of the 50 attempts passes all checks are excluded from the results table; they are not reported even as a fallback, because the highest-likelihood attempt for such specifications is typically a degenerate solution (e.g., a collapsed regime variance that inflates the log-likelihood artificially).

Each model is selected from 50 starts; only estimates that passed all checks (optimizer convergence, AR stationarity, parameter bounds, and valid transition probabilities) are reported. Specifications where no start passed all checks are excluded. Student-t degrees of freedom (ν) are held common across regimes; `Sw.Dist` reflects whether σ^2 is regime-specific.

Mean	Dist	K	Sw.AR	Sw.SPF	Sw.Dist	LogLik	AIC	BIC
Constant	Normal	2	N	N	Y	−186.6300	381.2600	394.7425
Constant	Normal	3	N	N	Y	−180.6569	379.3139	409.6496
Constant	Student t	2	N	N	Y	−185.7189	381.4378	398.2910
Constant	Student t	3	N	N	Y	−180.6407	381.2814	414.9877
ARX(1,1)	Normal	2	N	N	Y	−177.1889	368.3779	391.9723
ARX(1,1)	Normal	2	Y	N	Y	−175.1175	368.2350	398.5708
ARX(1,1)	Normal	2	Y	Y	Y	−175.0924	370.1848	403.8912
ARX(1,1)	Normal	3	N	N	Y	−174.3093	372.6186	413.0662
ARX(1,1)	Normal	3	Y	N	Y	−162.3751	356.7503	410.6805
ARX(1,1)	Student t	2	N	N	Y	−175.5345	367.0689	394.0340
ARX(1,1)	Student t	2	Y	N	Y	−166.6098	353.2196	386.9259
ARX(1,1)	Student t	2	Y	Y	Y	−166.4669	354.9338	392.0108
ARX(1,1)	Student t	3	Y	N	Y	−159.0882	352.1764	409.4772
ARX(2,1)	Normal	2	N	N	Y	−176.2156	368.4311	395.3962
ARX(2,1)	Normal	2	Y	N	Y	−174.9422	371.8845	408.9615
ARX(2,1)	Normal	2	Y	Y	Y	−174.3768	372.7535	413.2012
ARX(2,1)	Normal	3	Y	N	Y	−157.6865	353.3729	417.4150
ARX(2,1)	Student t	2	N	N	Y	−173.4411	364.8823	395.2180
ARX(2,1)	Student t	2	Y	N	Y	−162.6866	349.3733	389.8209
ARX(2,1)	Student t	2	Y	Y	Y	−161.4838	348.9676	392.7859
ARX(2,1)	Student t	3	Y	N	Y	−147.8112	335.6223	403.0351
ARX(2,2)	Normal	2	N	N	Y	−175.9610	369.9221	400.2578
ARX(2,2)	Normal	2	Y	N	Y	−174.5337	373.0674	413.5150
ARX(2,2)	Normal	2	Y	Y	Y	−174.1552	376.3105	423.4994
ARX(2,2)	Student t	2	N	N	Y	−173.2563	366.5127	400.2190
ARX(2,2)	Student t	2	Y	N	Y	−162.6810	351.3619	395.1802
ARX(2,2)	Student t	2	Y	Y	Y	−159.4737	348.9474	399.5070
ARX(2,2)	Student t	3	N	N	Y	−173.2563	376.5126	427.0722

6. Best Regime-Switching Model

Selected by AIC among estimates that passed optimizer convergence, stationarity, parameter-bound, distribution, and transition checks.

6.a. *Model Summary*

Mean process	Regime-switching parts	Non-switching parts	LogLik	AIC	BIC
ARX(2,1)	AR block, including c, shock distribution (σ^2, ν)	SPF block	−147.8112	335.6223	403.0351

6.b. *Switching Blocks*

Block	Switching
AR block, including c	Y
SPF block	N
Shock distribution (σ^2, ν)	Y

6.c. *Mean Process*

$$\pi_{t+1} = c_{s_t} + \rho_{1,s_t}\,\pi_t + \rho_{2,s_t}\,\pi_{t-1} + \phi_1\,SPF_t + u_{t+1} \tag{41}$$

6.d. *Mean Parameter Values*

Parameter	Switching	Regime 0	Regime 1	Regime 2
c	Y	0.4924	−0.0159	−0.3075
ρ_1	Y	−0.4103	0.1240	0.8039
ρ_2	Y	−0.1518	0.3155	−0.5983
ϕ_1	N	0.8977	0.8977	0.8977

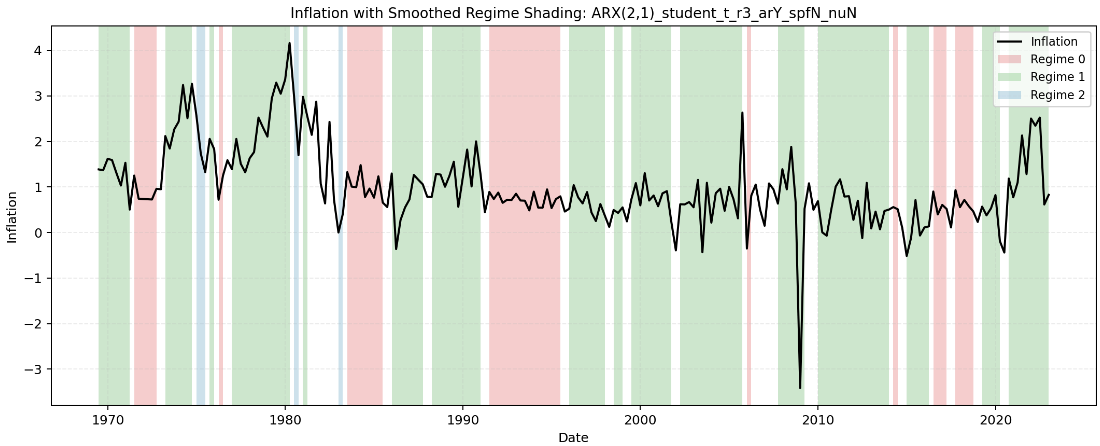
6.e. *Distribution Parameters*

Parameter	Switching	Regime 0	Regime 1	Regime 2
σ^2	Y	0.0282	0.3598	0.0174
ν	N	4.2263	4.2263	4.2263

6.f. *Transition Probability Matrix*

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.7435	0.0003	0.2562	1.0000
Regime 1	0.0648	0.8529	0.0823	1.0000
Regime 2	0.1517	0.6759	0.1724	1.0000

6.g. *Regime Classification Plot*


Inflation with Smoothed Regime Shading: ARX(2,1)_student_t_r3_arY_spfN_nuN
Inflation
Regime 0
Regime 1
Regime 2
Inflation
Date
1970
1980
1990
2000
2010
2020
4
3
2
1
0
−1
−2
−3

7. BEGE GARCH Family

All BEGE specifications use the same effective sample, residual definition, mean-process menu, random-search protocol, density evaluator, and result selection rules. The model-specific shape recursions and restrictions are reported.

7.a. Mean Processes

The BEGE exercises estimate all four mean processes:

- **Constant**: $\pi_{t+1} = SPF_t + u_{t+1}$.
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + u_{t+1}$.
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + u_{t+1}$.
- **ARX(2,2)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_{t+1}$.

The mean-parameter bounds used in estimation and collection are:

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	–	–	–	–	–
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$[-1, 1]$	–	$[-10, 10]$	–
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$[-2, 2]$	$[-1, 1]$	$[-10, 10]$	–
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$[-2, 2]$	$[-1, 1]$	$[-10, 10]$	$[-10, 10]$

Best-model collection also imposes mean-process stationarity before a BEGE estimate is eligible for reporting. For ARX(1,1), this requires $\mid \rho_1 \mid < 1$. For ARX(2,1) and ARX(2,2), the stationarity condition is the usual AR(2) polynomial restriction:

$$1 - \rho_1 z - \rho_2 z^2 = 0 \tag{42}$$

must have both roots outside the unit circle. Equivalently, the reported estimate must satisfy

$$\rho_1 + \rho_2 < 1, \qquad \rho_2 - \rho_1 < 1, \qquad \rho_2 > -1. \tag{43}$$

Only the inflation-lag coefficients ρ_1 and ρ_2 enter this stationarity check; the SPF coefficients ϕ_1 and ϕ_2 are treated as exogenous-regressor loadings and are not included in the AR polynomial.

Starting values for the estimated mean parameters are drawn uniformly from OLS-centered intervals, $\hat{\theta}_j \pm 2\, SE\left(\hat{\theta}_j\right)$.

7.b. Residual Dynamics

Each BEGE model writes the inflation residual as

$$u_t = \sigma_p \omega_{p,t} - \sigma_n \omega_{n,t}, \qquad \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \quad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1), \tag{44}$$

where $\tilde{\Gamma}(a, 1)$ is a centered gamma shock with shape a and unit scale. Conditional on p_t and n_t,

- conditional variance: $\sigma_p^2 p_t + \sigma_n^2 n_t$,
- conditional skewness numerator: $2\left(\sigma_p^3 p_t - \sigma_n^3 n_t\right)$,
- conditional excess-kurtosis numerator: $6\left(\sigma_p^4 p_t + \sigma_n^4 n_t\right)$.

7.c. BEGE Volatility Specifications

The common parameter bounds used in estimation and result collection are:

Parameter group	Bound
Fixed shapes $\bar{p}, \bar{n}$ and intercepts p_0, n_0	$[0, 10]$
Persistence parameters ρ, ρ_p, ρ_n	$[0, 1]$
Shock-loading parameters $\phi^{\pm}, \phi_p, \phi_n, \phi_p^{\pm}, \phi_n^{\pm}$	$[0, 2]$
Scale parameters σ_p, σ_n	$[10^{-5}, 2]$

To rule out numerically degenerate and explosive variance paths, I use an EWMA path as a scale-adaptive reference for the BEGE implied variance. This is a loose numerical safeguard, not a variance-targeting restriction.[3] For residuals u_t, define

$$EWMA_1 = \frac{\sum_{j=1}^{\tau} \lambda^{j-1} u_j^2}{\sum_{j=1}^{\tau} \lambda^{j-1}}, \qquad EWMA_t = \lambda EWMA_{t-1} + (1-\lambda) u_{t-1}^2, \quad t \geq 2. \tag{45}$$

The EWMA implied-variance bounds are imposed during optimization and checked again during collection. With $\lambda = 0.94$ and $\tau = \min(75, T)$, let $s_u^2 := 1/T \sum_{j=1}^{T} u_j^2$ and $M_u := 1 + \max_{1 \leq s \leq T} u_s^2$. Then

$$\begin{aligned} \underline{v}_t &:= \max\{10^{-6} EWMA_t,\ 10^{-8} s_u^2\}, \\ \overline{v}_t &:= \max\{\min(10^6 EWMA_t,\ 10^7 M_u),\ M_u\}, \\ \underline{v}_t &\leq \sigma_p^2 p_t + \sigma_n^2 n_t \leq \overline{v}_t, \qquad t = 1, \dots, T. \end{aligned} \tag{46}$$

7.c.i. *Constant BEGE:*

The Constant BEGE model keeps both shape parameters fixed:

$$p_t = \bar{p}, \qquad n_t = \bar{n}. \tag{47}$$

Thus $u_t = \sigma_p \omega_{p,t} - \sigma_n \omega_{n,t}$ with $\omega_{p,t} \sim \tilde{\Gamma}(\bar{p}, 1)$ and $\omega_{n,t} \sim \tilde{\Gamma}(\bar{n}, 1)$.

7.c.ii. *Symmetric BEGE:*

The Symmetric BEGE model uses the same persistence and shock-loading parameters in both shape recursions:

$$\begin{aligned} p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_n^2} (u_{t-1}^-)^2. \end{aligned} \tag{48}$$

The stability restriction is

$$\rho + \frac{\phi^+ + \phi^-}{2} < 1. \tag{49}$$

[3]The exponential weights follow the variance-forecasting idea in J.P. Morgan/Reuters, *RiskMetrics—Technical Document*, Fourth Edition (1996), Chapter 5, where recent squared shocks receive more weight than older shocks. Here the EWMA path is only a smoothed proxy for the local scale of u_t^2, not a BEGE variance target. The loose envelope is adapted from the `arch` package's `VolatilityProcess.variance_bounds`: the lower bound prevents the implied BEGE variance from collapsing toward zero, while the upper bound screens out explosive variance paths that would make likelihood evaluation numerically unreliable.

7.c.iii. *BadGood BEGE:*

The BadGood BEGE model lets each shape process react symmetrically to squared residuals, with separate good- and bad-environment parameters:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} u_{t-1}^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} u_{t-1}^2. \end{aligned} \tag{50}$$

The stability restrictions are

$$\rho_p + \phi_p < 1, \qquad \rho_n + \phi_n < 1. \tag{51}$$

7.c.iv. *Inflation/Deflation BEGE:*

The Inflation/Deflation BEGE model lets the good-news shape respond only to positive residuals and the bad-news shape respond only to negative residuals:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{52}$$

The stability restrictions are

$$\rho_p + \frac{\phi_p^+}{2} < 1, \qquad \rho_n + \frac{\phi_n^-}{2} < 1. \tag{53}$$

7.c.v. *Constant-p Full BEGE:*

Constant-p Full BEGE fixes the good-news shape and keeps the unrestricted Full BEGE update for the bad-news shape:

$$\begin{aligned} p_t &= p_0, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{54}$$

The dynamic shape process satisfies

$$\rho_n + \frac{\phi_n^+ + \phi_n^-}{2} < 1. \tag{55}$$

7.c.vi. *Constant-n Full BEGE:*

Constant-n Full BEGE fixes the bad-news shape and keeps the unrestricted Full BEGE update for the good-news shape:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0. \end{aligned} \tag{56}$$

The dynamic shape process satisfies

$$\rho_p + \frac{\phi_p^+ + \phi_p^-}{2} < 1. \tag{57}$$

7.c.vii. *Full BEGE:*

The Full BEGE model allows each shape process to respond separately to positive and negative residuals:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2. \end{aligned} \tag{58}$$

The stability restrictions are

$$\rho_p + \frac{\phi_p^+ + \phi_p^-}{2} < 1, \qquad \rho_n + \frac{\phi_n^+ + \phi_n^-}{2} < 1. \tag{59}$$

7.d. *Random Search*

BEGE estimation uses 50 independent seed jobs for each volatility specification. Within each seed job, each mean process uses 40 saved random draws, and each draw runs 25 optimizer starts. Therefore each BEGE specification and mean-process pair uses

$$50 \times 40 \times 25 = 50,000 \tag{60}$$

optimizer starts.

For volatility parameters, starting values are sampled inside the documented bounds. Shape intercept starts p_0, n_0 are sampled from the positive part of $[0, 10]$, scale starts σ_p, σ_n from $[10^{-5}, 2]$, and persistence and shock-loading starts are drawn so the relevant stability restriction is satisfied before optimization begins.

When a dynamic BEGE recursion is evaluated, I initialize the pre-sample p or n state by the parameter-implied unconditional backcast, $p_0/\left(1 - \rho - \frac{1}{2}(\phi^+ + \phi^-)\right)$ or the corresponding restricted formula, with a small positive numerical floor.[4] Constant-p and Constant-n models set the constant shape path directly.

7.e. *Density Evaluation*

Likelihood evaluation uses the stabilized BEGE density implementation documented in the BEGE Density section. The current evaluator keeps the closed-form hypergeometric expression in stable regions, switches to a guarded saddlepoint approximation when the closed form becomes numerically fragile.

7.f. *Selection And Reporting*

For each BEGE specification, the reported best-model page contains only the likelihood-best admissible estimate across mean processes. The by-mean best rows remain available in the CSV outputs linked from each report. Selection requires successful optimizer convergence, mean-process stationarity, documented parameter and stability constraints, and EWMA implied-variance bounds.

[4]The numerical floor is 10^{-4}: if the initial backcast or a later recursion produces a smaller value p_t, n_t, the code uses 0.0001 instead.

Each report also gives empirical 5%, median, and 95% quantiles for p_t, n_t, $\mathrm{Var}_t(u_t)$, the skewness numerator, and the excess-kurtosis numerator.

7.g. Standard Errors

I compute standard errors for the reported best estimate. For each observation, I approximate the score—the first derivative of that observation's log likelihood with respect to the parameters—using centered finite differences. I approximate the Hessian—the matrix of second derivatives of the total negative log likelihood—using the four-point centered finite-difference method implemented by `statsmodels.tools.numdiff.approx_hess`.

The default covariance is the sandwich matrix

$$\hat{V}\left(\hat{\theta}\right) = \hat{H}^{-1}\left(\sum_{t=1}^{T} \hat{g}_t {\hat{g}_t}'\right)\hat{H}^{-1}, \tag{61}$$

where $\hat{H}$ is the Hessian of the negative log likelihood and $\hat{g}_t$ is the numerical score contribution for observation t. If the sandwich calculation is numerically unstable, the collector uses an observed-information or inverse-OPG fallback. Standard errors that cannot be reliably identified are reported as `NA`.

8. BEGE Density

The BEGE density $f(u_t \mid p_t, n_t, \sigma_p, \sigma_n)$ is the conditional density of one inflation residual given the two gamma shape parameters and two scale parameters. This section compares three implementations:

- numerical integration, which is slow but useful as an independent benchmark;
- Justin's closed-form hypergeometric implementation;
- my stabilized modification on Justin's implementation.

The main conclusion is that Justin's closed-form formula is correct in stable regions, but its direct numerical evaluation can fail when the shape parameters enter the transition range where the BEGE distribution is already close to a Gaussian law. In those cases, very large terms in the hypergeometric expression nearly cancel. Finite-precision evaluation can then create artificial pointwise log densities that are numerically impossible and can dominate the likelihood search.

8.a. *Executive Summary*

- For moderate shape values, the current implementation and Justin's analytic implementation agree to numerical precision.
- On the fixed-shape timing tests below, the current implementation is about 1.5 times faster than Justin's implementation and about 957 times faster than 5,000-grid numerical integration at the median across the five test cases.
- The old job-side density produced implausibly large likelihoods on some dynamic BEGE estimates. For example, one BadGood ARX(2,1) row had stored LogLik 4920.7089, while the stabilized density gives −722.4342 and the saddlepoint check gives −722.7255.
- At the pointwise level, one old high-likelihood row reported $\ell_t = 90.3648$ for an observation with conditional variance 42.9373. The stabilized, saddlepoint, and Fourier-inversion values are all about −2.8008. This identifies a numerical failure of the hypergeometric evaluation, not an economic feature of the BEGE recursion.

8.b. *What Changed In `BEGE_density.py`*

Issue	Justin implementation	Current implementation	Reason
Hypergeometric evaluation	Direct `scipy.special.hyperu` or scalar `mpmath.hyperu` fallback.	Vectorized SciPy where stable, selective high-precision fallback, and log-domain asymptotic fallbacks when hyperu is nonfinite.	Avoid paying high-precision cost for every observation while keeping stable exact values.

Issue	Justin implementation	Current implementation	Reason
Large-shape transition region	Uses the closed-form branch until a hard max-shape switch.	Guards exact values once $p_t + n_t \geq 40$, blends exact and saddlepoint between total shape 50 and 80, and replaces exact values when exact and saddlepoint differ by more than 2 log units.	The failure is driven by cancellation in total shape, and it can occur before a max-shape cutoff is reached.
Near-normal limit	No explicit analytic normal-limit rule.	Uses a Gaussian density when total shape is at least 500 and standardized skewness and excess kurtosis are both below 0.03.	Gamma shocks converge analytically to a normal law under variance-preserving shape rescaling.
Density continuity	Hard branch switches can create likelihood discontinuities.	Uses log-sum-exp blending in the transition band.	Keeps the objective smoother for optimization.
Failure diagnostics	High likelihoods can silently enter estimation results.	The collector recomputes stored estimates with the stabilized density and writes path/layer diagnostics to CSV.	Prevents stale or unstable density values from determining best models.

The current thresholds are deliberately conservative: exact hypergeometric values are still used when they agree with saddlepoint values, while suspicious large-shape exact values are replaced by the saddlepoint approximation.

I compare three fixed-shape BEGE density implementations on the ARX(1,1) residuals from the canonical effective sample. As a benchmark, the Gaussian OLS log-likelihood is −207.381. The residuals are mean-zero (0.00) with a standard deviation of 0.636326, a negative skewness of −1.047441 and a large excess kurtosis of 8.210174.

8.c. Shape Parameter Range

The range below uses eligible ARX(1,1) rows from the BadGood BEGE results only. For each saved parameter vector I recomputed the recursive shape paths on the common ARX(1,1) residuals, then summarized all observations and all eligible rows. The density comparison itself keeps the selected shape values constant across time.

Model	Rows	p q05	p median	p q95	n q05	n median	n q95	sigma_p med	sigma_n med
BadGood	994	1.4285	2.8130	6.4330	0.0714	0.3320	2.8342	0.2355	0.5555

8.d. Fixed-Shape Parameter Sets

The log likelihood and timing comparison uses the following five BadGood-based parameter sets. In each row, `sigma_p` and `sigma_n` are held at their BadGood medians.

Set	p	n	sigma_p	sigma_n
p q05, n median	1.428538	0.331970	0.235491	0.555532
p median, n median	2.813004	0.331970	0.235491	0.555532
p q95, n median	6.433015	0.331970	0.235491	0.555532
p median, n q05	2.813004	0.071355	0.235491	0.555532
p median, n q95	2.813004	2.834194	0.235491	0.555532

8.e. Log Likelihood Comparison

`BEGE_density.py` is the current implementation. `BEGE_density_Justin.py` is Justin's analytic formula, and the numerical columns use `BEGE_density_Numerical_Integration.py::loglikedgam_constant` at the stated grid size.

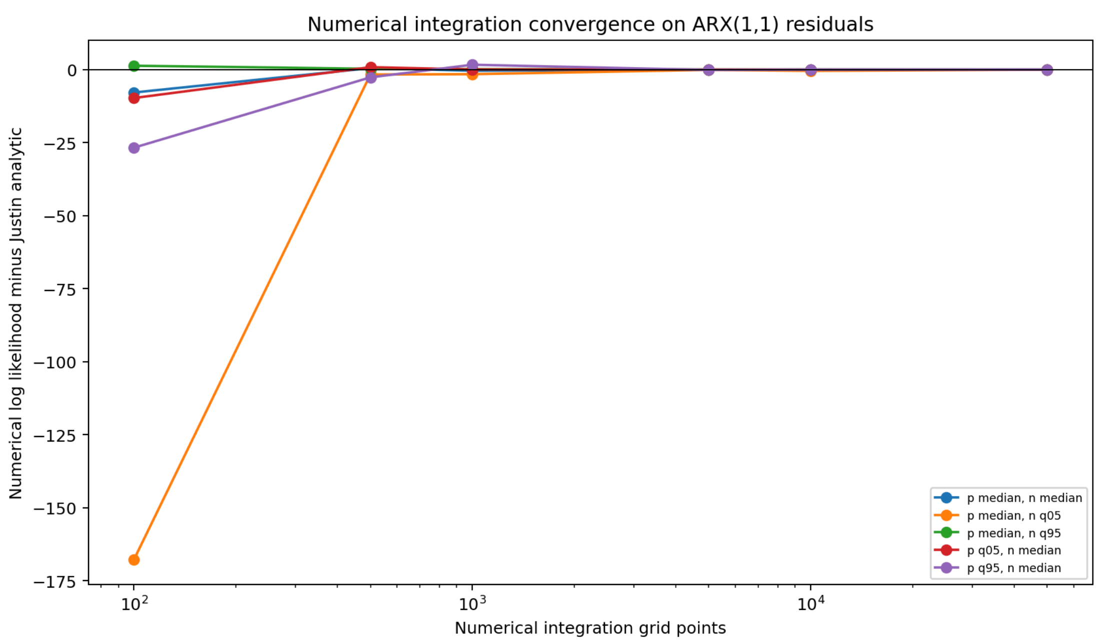
Parameter set	Current implementation	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	−214.162839	−214.162839	−223.900355	−213.330306	−214.116656	−214.148282	−214.146613	−214.160080
p median, n median	−188.957117	−188.957117	−196.801762	−188.438308	−189.359597	−189.033173	−188.947127	−188.954400
p q95, n median	−194.340481	−194.340481	−221.076698	−196.996227	−192.679189	−194.464527	−194.368904	−194.347495
p median, n q05	−212.877473	−212.877473	−380.654962	−214.556763	−214.448495	−212.976377	−213.323667	−212.904257
p median, n q95	−235.887717	−235.887717	−234.563988	−235.609633	−235.748009	−235.859932	−235.873999	−235.885265

- The old numerical integration method converges toward Justin's likelihood function as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood.

8.f. Evaluation Speed Comparison

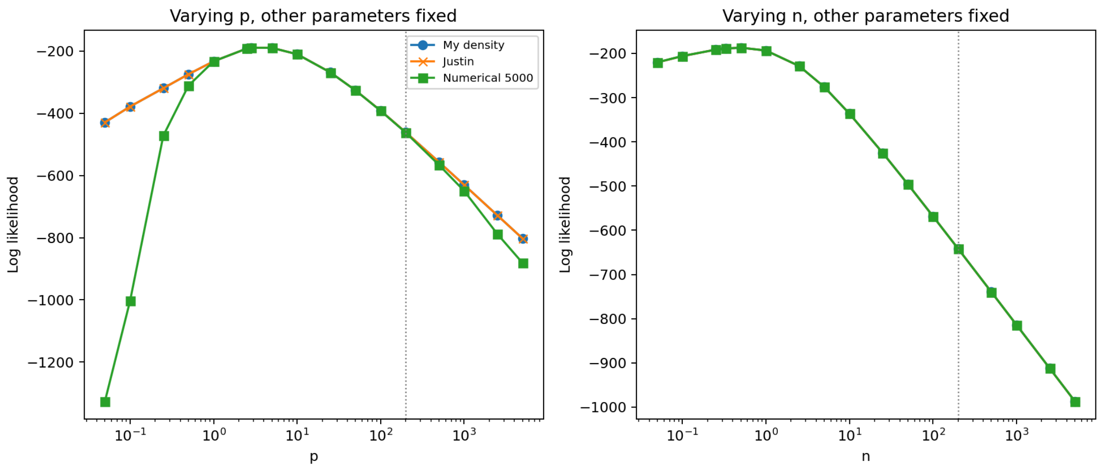
Parameter set	Current implementation	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	0.0003	0.0005	0.0086	0.0427	0.0883	0.4326	0.9102	4.5478
p median, n median	0.0004	0.0006	0.0075	0.0358	0.0730	0.3674	0.7613	3.8117
p q95, n median	0.0003	0.0005	0.0064	0.0308	0.0663	0.3258	0.6520	3.2415
p median, n q05	0.0003	0.0005	0.0076	0.0358	0.0737	0.3567	0.7479	3.6527
p median, n q95	0.0004	0.0006	0.0059	0.0284	0.0611	0.2900	0.5937	2.9724

- The current stabilized analytic implementation is slightly faster than Justin's direct analytic implementation in these moderate-shape cases (median speedup about 1.5 times).
- The current implementation is orders of magnitude faster than numerical integration. Relative to the 5,000-grid integration line, the median speedup is about 957 times across the five parameter sets.



8.g. Shape Tail Consistency

Holding the other parameters at the BadGood medians (`p=2.8130`, `n=0.3320`, `sigma_p=0.2355`, `sigma_n=0.5555`), I vary either `p` or `n` up to 5000 and compare the three density functions. The numerical integration line uses 5,000 grid points.



8.h. Robust Large-Shape Evaluation

The closed-form BEGE density contains a confluent hypergeometric term. It is accurate for small and moderate shape states, but it can become numerically fragile when the recursive shapes are large because several log terms nearly cancel. In that region, a low-precision evaluation can create artificial pointwise log densities that are orders of magnitude too high or too low.

For estimation and result collection, `BEGE_density.py` now uses a guarded large-shape rule:

- keep the exact hypergeometric expression for small-shape observations;
- use the cumulant saddlepoint density when total shape enters the fragile range, with a smooth transition rather than a single discontinuous cutoff;
- replace exact values by the saddlepoint value whenever the two disagree by more than the numerical tolerance in the guarded region;
- use the Gaussian density when total shape is large and standardized skewness and excess kurtosis are both numerically close to zero.

The saddlepoint approximation is based on the conditional cumulant-generating function of

$$u_t = \sigma_p(\Gamma(p_t, 1) - p_t) - \sigma_n(\Gamma(n_t, 1) - n_t). \tag{62}$$

It solves $K'(\hat{s}) = u_t$ and evaluates

$$\log f(u_t) \approx K(\hat{s}) - \hat{s}u_t - \frac{1}{2}\log\{2\pi K''(\hat{s})\}. \tag{63}$$

This construction respects the analytic normal limit of the BEGE distribution: as the gamma shapes grow while $\sigma_p^2 p_t + \sigma_n^2 n_t$ remains fixed, the standardized skewness and excess kurtosis shrink toward zero and the large-shape density converges to the Gaussian density with the same conditional variance.

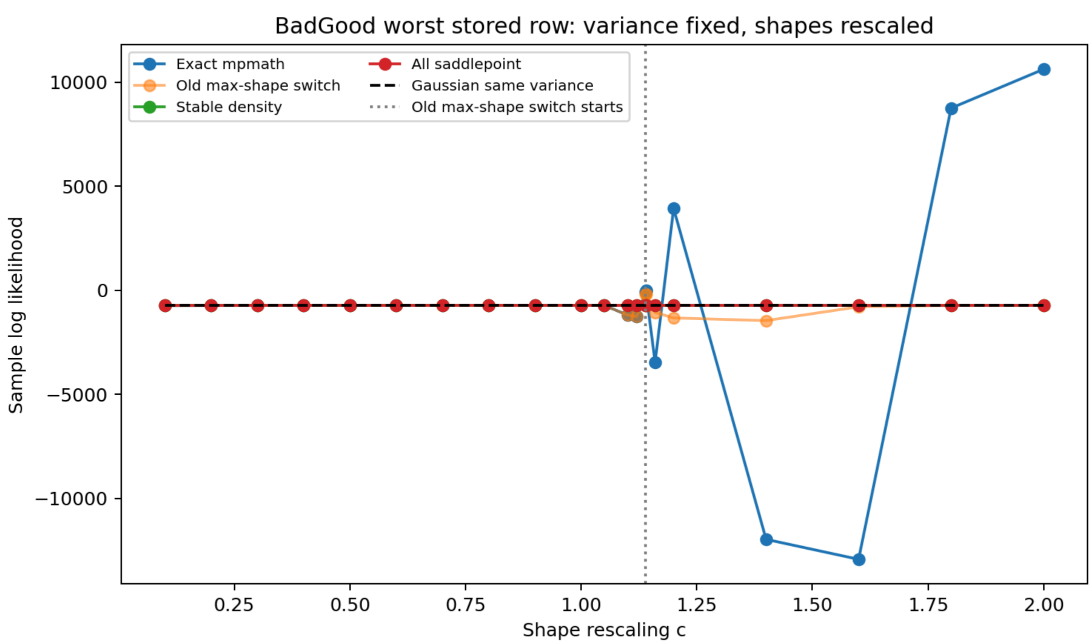
8.i. Saddlepoint Threshold Experiment

The BEGE distribution should approach a Gaussian law when both gamma shapes grow and the scale parameters shrink so that $\sigma_p^2 p_t + \sigma_n^2 n_t$ stays fixed. This check takes the same seed 45, draw 25, ARX(2,1) path and rescales

$$p_t(c) = cp_t, \qquad n_t(c) = cn_t, \qquad \sigma_p(c) = \sigma_p/\sqrt{c}, \qquad \sigma_n(c) = \sigma_n/\sqrt{c}. \tag{64}$$

The variance path is unchanged by construction. The exact hypergeometric branch is not monotonically better or worse as shapes grow: it is fragile in transition regions where large log terms cancel. A hard cutoff based only on `max(p_t,n_t)` is also too late. For this row, the old `max(p_t,n_t)=180` switch starts at $c = 1.1395$, while exact-branch failures already appear around $c = 1.10$ and the stored old job-side likelihood at $c = 1$ was 4920.7089.

Scale c	Median total shape	Max shape	Old max-shape saddle obs.	Exact mpmath LogLik	Old max-shape switch LogLik	Stabilized LogLik	Saddlepoint LogLik	Gaussian LogLik
0.1000	19.4893	15.7964	0	−728.2549	−728.2549	−728.2549	−727.1415	−720.5045
0.4000	77.9574	63.1856	0	−724.0184	−724.0184	−724.0388	−723.9614	−720.5045
0.8000	155.9147	126.3712	0	−722.6555	−722.6555	−722.6530	−722.9801	−720.5045
1.0000	194.8934	157.9640	0	−722.4380	−722.4380	−722.4342	−722.7255	−720.5045
1.1000	214.3827	173.7604	0	−1187.7835	−1187.7835	−722.3558	−722.6246	−720.5045
1.1395	222.0811	180.0000	1	−94.4486	−174.4248	−722.3280	−722.5884	−720.5045
1.2000	233.8721	189.5568	162	3934.4808	−1329.2867	−722.2885	−722.5364	−720.5045
1.8000	350.8081	284.3352	197	8749.6077	−722.0183	−722.0077	−722.1699	−720.5045
2.0000	389.7868	315.9280	200	10616.6236	−721.9500	−721.9408	−722.0858	−720.5045



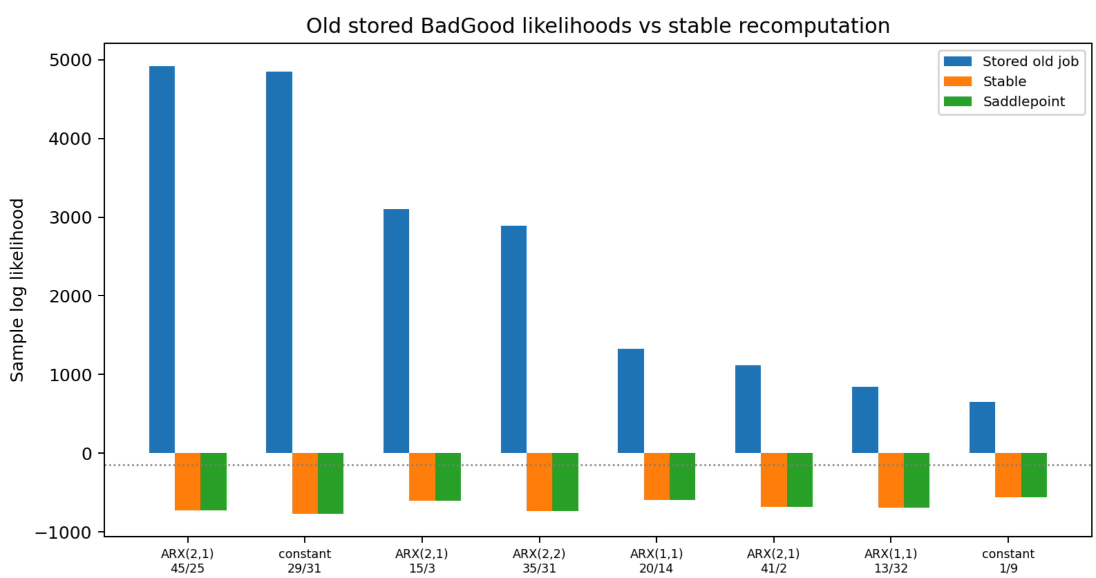
The implemented rule therefore uses total shape, not a hard maximum-shape cap:

- guard exact hypergeometric values once `p_t+n_t >= 40`;
- smooth-blend exact and saddlepoint values between total shape 50 and 80;
- replace exact values when exact and saddlepoint differ by more than 2 log units in the guarded region;
- use the Gaussian limit only after total shape 500 and only when standardized skewness and excess kurtosis are both below 0.03 in absolute/magnitude terms.

8.j. *Gigantic Stored Log-Likelihood Diagnostics*

The table below starts from the BadGood results saved by the old production jobs. `Stored job LogLik` is the likelihood written by that old job-side density evaluation. The same parameter vectors are then recomputed on the same effective-sample residuals and recursive shape paths using the stabilized density, all-saddlepoint density, Gaussian density with the same conditional variance path, and reproducible exact branches available in the current git history.

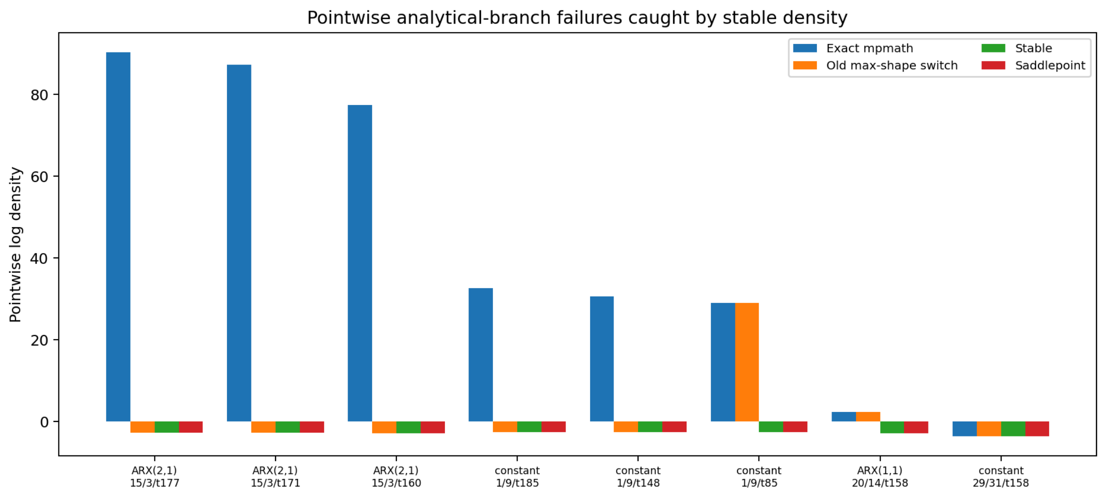
Mean	Seed	Draw	Stored job LogLik	Stabilized LogLik	Saddlepoint LogLik	Gaussian LogLik	Exact mpmath LogLik	Max shape	Median total shape	Median σ_t^2
ARX(2,1)	45	25	4920.7089	−722.4342	−722.7255	−720.5045	−722.4380	157.9640	194.8934	135.4737
constant	29	31	4846.4315	−770.7119	−770.6706	−770.2032	−1889.5667	171.9426	234.7388	225.1110
ARX(2,1)	15	3	3098.3394	−599.0093	−598.9692	−598.1718	−1186.8621	199.7122	220.0504	42.6849
ARX(2,2)	35	31	2892.0304	−734.4489	−734.2511	−730.1723	−807.8124	173.4795	205.7171	146.5852
ARX(1,1)	20	14	1330.8412	−590.1301	−590.0846	−592.7498	−584.8732	170.2031	206.8959	40.1486
ARX(2,1)	41	2	1113.7517	−684.6683	−684.6022	−680.7040	−684.8619	146.8413	164.9777	90.3592
ARX(1,1)	13	32	846.3034	−694.3325	−694.2827	−692.9410	−791.0434	156.0623	190.5299	104.1080
constant	1	9	651.3535	−561.1096	−561.0324	−561.6200	−1101.1363	209.9112	191.0321	30.1035



Across 8,000 BadGood rows in this diagnostic pass, the stored job-side likelihood produced 150 rows above `-150`, including 13 rows above zero. The stabilized recomputation produced zero rows above `-150`, and its largest value was −161.8929.

Some pointwise analytical-branch failures are reproducible from the current git history. The table reports the observations with the largest exact-vs-stable gaps among the old high-likelihood rows. The Fourier inversion column is a slow direct numerical check for the largest gaps.

Mean	Seed	Draw	Obs.	u_t	p_t	n_t	σ_t^2	Exact mpmath ℓ_t	Stabilized ℓ_t	Saddlepoint ℓ_t	Gaussian ℓ_t	Fourier ℓ_t
ARX(2,1)	15	3	177	0.0913	42.9229	187.7988	42.9373	90.3648	−2.8008	−2.8008	−2.7989	−2.8007
ARX(2,1)	15	3	171	0.2476	42.9393	189.9662	42.9916	87.2647	−2.8052	−2.8052	−2.8002	−2.8051
ARX(2,1)	15	3	160	0.1429	44.3305	199.7122	44.4494	77.4285	−2.8191	−2.8191	−2.8163	−2.8191
constant	1	9	185	−0.5080	188.8323	17.0463	30.2154	32.6685	−2.6480	−2.6480	−2.6274	−2.6503
constant	1	9	148	0.5209	180.3713	17.1093	30.2612	30.6176	−2.6069	−2.6069	−2.6284	−2.6090
constant	1	9	85	1.0855	174.4991	17.0414	30.1040	28.9611	−2.5957	−2.5957	−2.6408	−2.5979



These rows show why the huge stored likelihoods are not economically or numerically credible. A conditional variance around 30 to 44 cannot support pointwise log densities between 29 and 90. The stabilized values match saddlepoint and Fourier inversion, so the problem is numerical evaluation of the hypergeometric expression, not the BEGE recursion itself.

8.k. Findings

- BadGood quantiles provide the fixed-shape examples used in the likelihood and timing tables.
- The numerical integration function is useful as a convergence diagnostic, but it is much slower than the analytic functions and remains grid-sensitive.
- The shape-tail plot varies one shape parameter at a time while holding the other shape and both scale parameters at the BadGood medians.

- The saddlepoint switch should be based on guarded total shape and exact-vs-saddlepoint disagreement; `max(p_t,n_t)` is useful as a diagnostic but is not a reliable likelihood cutoff.
- Large-shape likelihood evaluation should not rely only on the closed-form hypergeometric expression. The guarded saddlepoint and Gaussian-limit checks prevent spurious gigantic likelihood values from determining BEGE estimates.

9. Constant BEGE Best Model Summary

Generated: `2026-06-18T10:10:04` Total estimations: `8000` Successful estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate across mean processes. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

9.a. *Selected Best Model*

Best admissible estimate ranked by log likelihood across mean processes.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	30	30	−181.6884	381.3768	411.7126

Empirical path quantiles:

Series	5%	Median	95%
p_t	2.6646	2.6646	2.6646
n_t	0.2821	0.2821	0.2821
σ_t^2	0.3952	0.3952	0.3952
s_t^2	−0.2286	−0.2286	−0.2286
k_t^2	0.9308	0.9308	0.9308

Mean process:

$$\pi_{t+1} = 0.0041 + 0.3232\,\pi_t + 0.1633\,\pi_{t-1} + 0.4107\,SPF_t + 0.1942\,SPF_{t-1} + u_{t+1} \tag{65}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.2712\,\omega_{p,t} - 0.8404\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(\bar{p}, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(\bar{n}, 1), \\ \bar{p} &= 2.6646, \qquad \bar{n} = 0.2821, \\ \mathrm{Var}_t(u_t) &= (0.2712)^2\,2.6646 + (0.8404)^2\,0.2821. \end{aligned} \tag{66}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.0041	0.1115
ρ_1	0.3232	0.1214

ρ_2	0.1633	0.0829
ϕ_1	0.4107	0.3014
ϕ_2	0.1942	0.2957
$\bar{p}$	2.6646	0.1839
$\bar{n}$	0.2821	0.1597
σ_p	0.2712	0.0218
σ_n	0.8404	0.3557

10. Symmetric BEGE Best Model Summary

Generated: `2026-06-18T10:10:15` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

10.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	35	12	−167.9012	359.8025	400.2501

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.6347	1.1647	5.6629
n_t	0.3243	0.5949	2.8917
σ_t^2	0.1638	0.3005	1.4610
s_t^2	−0.2097	−0.0432	−0.0235
k_t^2	0.1876	0.3441	1.6727

Mean process:

$$\pi_{t+1} = 0.2006 + 0.1978\,\pi_t + 0.2405\,\pi_{t-1} + 0.2257\,SPF_t + 0.1197\,SPF_{t-1} + u_{t+1} \tag{67}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3592\,\omega_{p,t} - 0.5026\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{68}$$

$$\begin{aligned} p_t &= 0.1762 + 0.6852\,p_{t-1} + \frac{0.4128}{2(0.3592)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0809}{2(0.3592)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.0900 + 0.6852\,n_{t-1} + \frac{0.4128}{2(0.5026)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0809}{2(0.5026)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{69}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.2006	0.0397
ρ_1	0.1978	0.0273
ρ_2	0.2405	0.0389
ϕ_1	0.2257	0.2154
ϕ_2	0.1197	0.1590
p_0	0.1762	0.0688
n_0	0.0900	0.0393
ρ	0.6852	0.0587
ϕ^+	0.4128	0.0560
ϕ^-	0.0809	0.0871
σ_p	0.3592	0.0670
σ_n	0.5026	0.0621

11. BadGood BEGE Best Model Summary

Generated: `2026-06-18T10:10:22` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

11.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,1)	39	14	−163.5594	351.1188	391.5664

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.4097	0.5799	1.7285
n_t	0.2985	0.7108	3.8717
σ_t^2	0.1416	0.2554	1.0320
s_t^2	−0.2196	0.0384	0.0683
k_t^2	0.1754	0.2991	1.1456

Mean process:

$$\pi_{t+1} = 0.2119 + 0.2240\,\pi_t + 0.2742\,\pi_{t-1} + 0.2699\,SPF_t + u_{t+1} \tag{70}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.4778\,\omega_{p,t} - 0.4042\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{71}$$

$$\begin{gathered} p_t = 0.1575 + 0.5795\,p_{t-1} + \frac{0.2143}{2(0.4778)^2}\,u_{t-1}^2, \\ n_t = 0.1900 + 0.2831\,n_{t-1} + \frac{0.6633}{2(0.4042)^2}\,u_{t-1}^2 \end{gathered} \tag{72}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.2119	0.0283
ρ_1	0.2240	0.0281
ρ_2	0.2742	0.0140
ϕ_1	0.2699	0.0447
p_0	0.1575	0.0329
n_0	0.1900	0.0469
ρ_p	0.5795	0.0474
ρ_n	0.2831	0.0315
ϕ_p	0.2143	0.0433
ϕ_n	0.6633	0.0058
σ_p	0.4778	0.0509
σ_n	0.4042	0.0328

12. Inflation/Deflation BEGE-GJR Best Model Summary

Generated: `2026-06-18T10:11:34` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

12.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	3	39	−167.9018	361.8035	405.6218

Empirical path quantiles:

Series	5%	Median	95%
p_t	4.5793	7.2032	28.6664
n_t	0.0584	0.0697	0.6744
σ_t^2	0.1699	0.2766	1.0806
s_t^2	−1.2028	−0.0927	0.0660
k_t^2	0.3575	0.4378	3.8913

Mean process:

$$\pi_{t+1} = 0.1246 + 0.2617\,\pi_t + 0.1743\,\pi_{t-1} + 0.2915\,SPF_t + 0.1749\,SPF_{t-1} + u_{t+1} \tag{73}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.1486\,\omega_{p,t} - 0.9893\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{74}$$

$$\begin{aligned} p_t &= 1.2658 + 0.7076\,p_{t-1} + \frac{0.4250}{2(0.1486)^2}\left(u_{t-1}^{+}\right)^2, \\ n_t &= 0.0508 + 0.1317\,n_{t-1} + \frac{1.2626}{2(0.9893)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{75}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.1246	0.0726
ρ_1	0.2617	0.0767
ρ_2	0.1743	0.0758
ϕ_1	0.2915	0.3736
ϕ_2	0.1749	0.3151
p_0	1.2658	0.7514
n_0	0.0508	0.0486
ρ_p	0.7076	0.1141
ρ_n	0.1317	0.0583
ϕ_p^+	0.4250	0.1637
ϕ_n^-	1.2626	0.8901
σ_p	0.1486	0.0423
σ_n	0.9893	0.5903

13. Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:22` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

13.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	13	33	−165.2805	360.5611	411.1206

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.9806	1.1858	2.3678
n_t	0.0849	0.2556	2.8363
σ_t^2	0.1834	0.2969	1.1546
s_t^2	−0.9826	0.0427	0.1478
k_t^2	0.1925	0.3661	2.0843

Mean process:

$$\pi_{t+1} = 0.1937 + 0.1415\,\pi_t + 0.1380\,\pi_{t-1} + 0.2283\,SPF_t + 0.3635\,SPF_{t-1} + u_{t+1} \tag{76}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3802\,\omega_{p,t} - 0.5787\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{77}$$

$$\begin{aligned} p_t &= 0.0347 + 0.9561\,p_{t-1} + \frac{0.0000}{2(0.3802)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0334}{2(0.3802)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.0295 + 0.5611\,n_{t-1} + \frac{0.7784}{2(0.5787)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0720}{2(0.5787)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{78}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.1937	0.0783
ρ_1	0.1415	0.0611
ρ_2	0.1380	0.0618
ϕ_1	0.2283	0.2319
ϕ_2	0.3635	0.1626
p_0	0.0347	0.0277
n_0	0.0295	0.0258
ρ_p	0.9561	0.0242
ρ_n	0.5611	0.0879
ϕ_p^+	0.0000	NA
ϕ_p^-	0.0334	0.0159
ϕ_n^+	0.7784	0.1702
ϕ_n^-	0.0720	0.0947
σ_p	0.3802	0.0621
σ_n	0.5787	0.1108

14. Constant-p Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:11` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

14.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	28	25	−166.7909	357.5819	398.0296

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.4727	0.4727	0.4727
n_t	0.4590	0.9268	7.3165
σ_t^2	0.1990	0.2674	1.2023
s_t^2	−0.6797	0.0355	0.0878
k_t^2	0.2795	0.3396	1.1602

Mean process:

$$\pi_{t+1} = 0.1831 + 0.1660\,\pi_t + 0.0787\,\pi_{t-1} + 0.2999\,SPF_t + 0.3076\,SPF_{t-1} + u_{t+1} \tag{79}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.5281\,\omega_{p,t} - 0.3825\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{80}$$

$$\begin{aligned} p_t &= 0.4727, \\ n_t &= 0.2113 + 0.4747\,n_{t-1} + \frac{0.7577}{2(0.3825)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.2362}{2(0.3825)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{81}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.1831	0.0290

ρ_1	0.1660	0.0376
ρ_2	0.0787	0.0273
ϕ_1	0.2999	0.1631
ϕ_2	0.3076	0.1028
p_0	0.4727	0.1424
n_0	0.2113	0.0517
ρ_n	0.4747	0.0291
ϕ_n^+	0.7577	0.0098
ϕ_n^-	0.2362	0.0348
σ_p	0.5281	0.1365
σ_n	0.3825	0.0364

15. Constant-n Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:30` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

15.a. *Selected Best Model*

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	31	40	−170.2704	364.5408	404.9885

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.6223	1.1422	5.6821
n_t	0.4055	0.4055	0.4055
σ_t^2	0.2157	0.2840	0.8798
s_t^2	−0.0950	−0.0455	0.3862
k_t^2	0.3302	0.3840	0.8532

Mean process:

$$\pi_{t+1} = 0.1460 + 0.1849\,\pi_t + 0.0812\,\pi_{t-1} + 0.9787\,SPF_t + -0.3699\,SPF_{t-1} + u_{t+1} \tag{82}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3623\,\omega_{p,t} - 0.5750\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{83}$$

$$\begin{aligned} p_t &= 0.1443 + 0.7377\,p_{t-1} + \frac{0.4074}{2(0.3623)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0184}{2(0.3623)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.4055 \end{aligned} \tag{84}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.1460	0.0336

ρ_1	0.1849	0.0432
ρ_2	0.0812	0.0439
ϕ_1	0.9787	0.1048
ϕ_2	−0.3699	0.0891
p_0	0.1443	0.0321
n_0	0.4055	0.0525
ρ_p	0.7377	0.0172
ϕ_p^+	0.4074	0.0285
ϕ_p^-	0.0184	0.0323
σ_p	0.3623	0.0408
σ_n	0.5750	0.0736